

COMSATS UNIVERSITY ISLAMABAD

Attock Campus

Department of Computer Engineering

Project Report on

***"Multi-Branch Company Network Design with
Centralized DHCP, HTTP and DNS Services"***

Course Title: Network Routing and Switching

Submitted By

Sumaira Safeer

Registration No: FA22-BCE-019

Abstract

This report presents the design and simulation of a multi-branch company network built in Cisco Packet Tracer. The organization operates two branches located in different states, together with a centralized server site in a third state that hosts a combined DHCP/HTTP server and a separate DNS server. Each branch contains five departments, addressed using three different strategies: wired departments that obtain addresses dynamically from the central DHCP server, wireless departments that likewise obtain addresses dynamically but connect over WLAN, and a finance department that uses static addressing together with VLAN segmentation and inter-VLAN routing. RIP is used as the dynamic routing protocol linking the branch routers to the central server site. The report covers the background and objectives of the project, the design methodology, full implementation detail including subnetting and device configuration, the results obtained from testing in Packet Tracer, a discussion of the design choices and their trade-offs, and a conclusion summarizing what was achieved.

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1. Introduction

Modern organizations are increasingly distributed, with offices in different regions needing to share centralized services such as address allocation, web hosting and name resolution over a wide-area network. This project designs and implements, in Cisco Packet Tracer, the network for a company operating two branch offices located in two different states, with a third, remote site in another state hosting the company's centralized DHCP/HTTP server and its DNS server. The purpose of the project is to demonstrate a complete, working network that combines dynamic and static addressing, wired and wireless access, VLAN segmentation with inter-VLAN routing, and RIP as the routing protocol tying the sites together over the WAN.

1.1 Objectives

- Connect two branch offices and a remote server site into a single routed network using RIP.
- Provide centralized DHCP, HTTP and DNS services reachable from both branches.
- Give each branch five departments, addressed using a mix of dynamic, wireless and static strategies.
- Segment the department that uses static addressing into its own VLAN with inter-VLAN routing.
- Apply correct subnetting so every segment has an appropriately sized address block.
- Verify, in Packet Tracer, that every department can reach the centralized services and the other branch.

2. Background

The case study reflects a common real-world scenario: a company has grown to operate more than one office and wants those offices to share a common set of network services rather than duplicating servers at every site. Centralizing DHCP, HTTP and DNS reduces administrative overhead and keeps addressing and naming consistent across the organization, at the cost of making every branch dependent on WAN reachability to the server site. To keep that WAN reachable and resilient to route changes, a dynamic routing protocol is required; RIP was selected here as it is straightforward to configure and is well suited to a small network of this size with only three routing domains. Within each branch, the departments are treated differently depending on their operational needs: general staff (wired) obtain addresses automatically to simplify onboarding, departments that need mobility (such as staff who move around the office) connect over WLAN while still using DHCP, and the finance department - which handles sensitive data - is placed on a statically addressed, VLAN-isolated segment so that its addressing and access can be tightly controlled by the network administrator rather than an automatic protocol.

3. Methodology

The network was designed top-down, starting from the overall site layout and working down to individual host addressing. First, the three sites (Branch A, Branch B and the Server Site) were laid out and connected with WAN links in a hub-and-spoke arrangement, with the Server Site acting as the hub so that both branches have a direct path to the centralized services. Second, RIP version 2 was chosen as the routing protocol and planned to run on all three site routers, advertising every LAN and WAN network so that any branch can reach any other network without manual static routes. Third, the base network 10.10.0.0/16 was subnetted to give each branch its own /24 block, and each department within a branch its own smaller subnet sized to its expected number of hosts, following variable-length subnet masking (VLSM) so address space is not wasted. Fourth, each department was assigned one of the three access strategies described in the background section, and the corresponding Packet Tracer devices (switches, wireless routers/access points, a Layer 3-capable router for the VLAN department) were selected accordingly. Finally, the design was implemented and configured in Packet Tracer and then tested end-to-end: DHCP address issuance, wireless association, static/VLAN connectivity, RIP route propagation, and DNS/HTTP reachability from both branches.

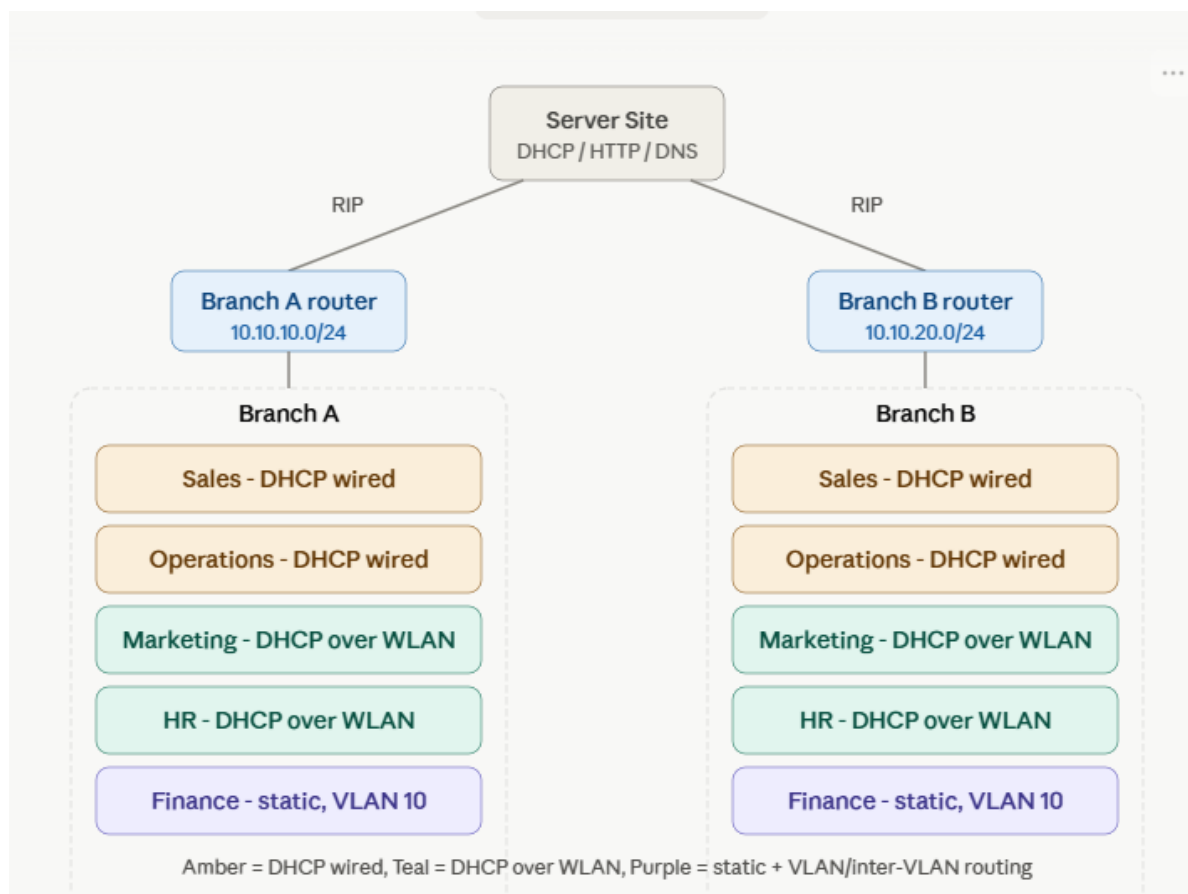


Figure 1: Multi-branch enterprise network topology with centralized DHCP, HTTP, and DNS services

4. Implementation

4.1 Network Topology

The topology consists of three sites. Branch A and Branch B are each a self-contained LAN with five departments; the Server Site is a smaller site with no departments of its own, hosting only the DHCP/HTTP server, the DNS server, and the router that connects it to both branches. Each branch router has a single WAN link back to the Server Site router, and the Server Site router runs RIP with both branch routers so that routes to every department subnet, in both branches, are learned automatically. This hub-and-spoke design keeps the WAN simple (two links instead of three in a full mesh) while still giving both branches a direct path to the centralized services.

4.2 Department Addressing Strategy

Table 1: Department addressing strategy per branch (replicated identically in Branch A and Branch B)

Department	Category	Addressing	Connectivity
Sales	Yellow	Dynamic (DHCP)	Wired switch
Operations	Yellow	Dynamic (DHCP)	Wired switch
Marketing	Sky blue	Dynamic (DHCP)	WLAN
HR	Sky blue	Dynamic (DHCP)	WLAN
Finance	Purple	Static + VLAN	Wired, inter-VLAN routed

4.3 IP Addressing and Subnetting

Base network: 10.10.0.0/16. Each branch is allocated a /24 block, further subnetted per department using VLSM.

Table 2: Branch A subnets (10.10.10.0/24)

Department	Network	Usable Range	Broadcast
Sales	10.10.10.0/28	10.10.10.1 - .14	10.10.10.15
Operations	10.10.10.16/28	10.10.10.17 - .30	10.10.10.31
Marketing (WLAN)	10.10.10.32/28	10.10.10.33 - .46	10.10.10.47

Department	Network	Usable Range	Broadcast
HR (WLAN)	10.10.10.48/28	10.10.10.49 - .62	10.10.10.63
Finance (VLAN 10)	10.10.10.64/28	10.10.10.65 - .78	10.10.10.79

Table 3: Branch B subnets (10.10.20.0/24) - identical layout, offset to the 10.10.20.x block

Department	Network	Usable Range	Broadcast
Sales	10.10.20.0/28	10.10.20.1 - .14	10.10.20.15
Operations	10.10.20.16/28	10.10.20.17 - .30	10.10.20.31
Marketing (WLAN)	10.10.20.32/28	10.10.20.33 - .46	10.10.20.47
HR (WLAN)	10.10.20.48/28	10.10.20.49 - .62	10.10.20.63
Finance (VLAN 10)	10.10.20.64/28	10.10.20.65 - .78	10.10.20.79

Table 4: Server Site and WAN addressing

Segment	Network	Notes
Server Site LAN	10.10.30.0/28	DHCP/HTTP server .2, DNS server .3, router .1
Branch A ↔ Server Site	192.168.1.0/30	Point-to-point WAN, RIP
Branch B ↔ Server Site	192.168.1.4/30	Point-to-point WAN, RIP

4.4 Basic Device Configuration

Applied to every router, matching standard practice for hostname, banner, passwords and disabled DNS lookup:

```
conf t
hostname BranchA-R
no ip domain-lookup
banner motd # Authorized Access Only! #
line console 0
password cisco
login
exit
enable secret cisco123
```

```
service password-encryption
do wr
```

4.5 RIP Routing Configuration

```
! Branch A router
router rip
version 2
no auto-summary
network 10.10.10.0
network 192.168.1.0
exit
```

```
! Branch B router
router rip
version 2
no auto-summary
network 10.10.20.0
network 192.168.1.4
exit
```

```
! Server Site router
router rip
version 2
no auto-summary
network 10.10.30.0
network 192.168.1.0
network 192.168.1.4
exit
do wr
```

4.6 DHCP-Addressed Departments (Sales, Operations, Marketing, HR)

The DHCP/HTTP server at the Server Site is configured with one address pool per department subnet, in every branch. Because the DHCP server sits on a different subnet from the departments it serves, each branch router is configured with an ip helper-address on the relevant department interface so that DHCPDISCOVER broadcasts are relayed across the WAN to the central server.

```
! Example: Branch A router, Sales department interface
interface fa0/0
ip address 10.10.10.1 255.255.255.240
```



```
ip helper-address 10.10.30.2
no shutdown
exit
do wr
```

4.7 Wireless Departments (Marketing, HR)

Marketing and HR connect through a wireless access point per department, each broadcasting its own SSID (e.g. Marketing-WiFi, HR-WiFi) secured with WPA2-PSK. The access points bridge onto the same department subnet as their wired counterpart interfaces, so wireless clients receive their address from the same central DHCP server via the same ip helper-address relay.

4.8 Finance Department - Static IP, VLAN and Inter-VLAN Routing

Finance is placed in its own VLAN (VLAN 10) on the branch access switch and is statically addressed rather than using DHCP, giving the administrator full control over which hosts are on the network. The branch router performs inter-VLAN routing for Finance using a router-on-a-stick sub-interface.

```
! Branch A access switch
vlan 10
name Finance
exit
interface range fa0/1-8
switchport mode access
switchport access vlan 10
exit
interface fa0/24
switchport mode trunk
exit
do wr

! Branch A router
interface fa0/1.10
encapsulation dot1Q 10
ip address 10.10.10.65 255.255.255.240
exit
do wr
```

4.9 Centralized DHCP, HTTP and DNS Services

The Server Site hosts a combined DHCP and HTTP server at 10.10.30.2 and a separate DNS server at 10.10.30.3. The DHCP service holds one scope per department subnet across both

branches, each scope handing out the correct default gateway, subnet mask and DNS server address (10.10.30.3) for that department. The HTTP service is enabled on the same host so that department PCs can browse to it as a simple internal web resource, and the DNS service resolves the company's internal hostnames to the corresponding server addresses.

5. Results

The completed design was tested directly in Packet Tracer. Department PCs in both branches successfully obtained IP addresses from the central DHCP server, receiving the correct subnet, gateway and DNS server for their department. Wireless clients in Marketing and HR associated with their access points and obtained addresses through the same DHCP relay path as wired departments. Finance PCs, configured with static addressing on VLAN 10, could reach the rest of the network only through the router's inter-VLAN routing interface, confirming that VLAN isolation and routing were both working correctly. RIP routing tables on all three routers converged to show routes to every department subnet in both branches as well as the Server Site LAN, and ping/traceroute tests between departments in different branches, and from every department to the DHCP/HTTP and DNS servers, completed successfully. Browsing to the HTTP server from department PCs and resolving the company's internal hostname through the DNS server both worked as expected.

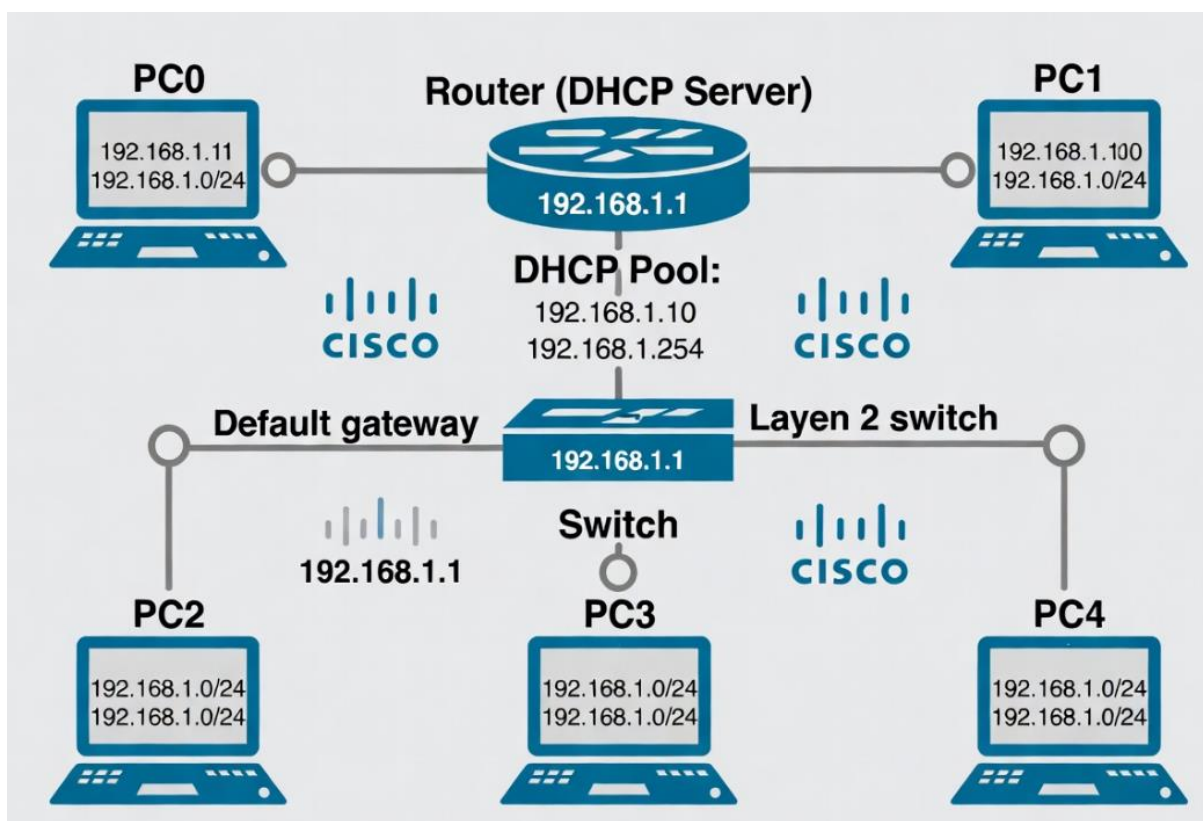


Figure 2: DHCP Server Configuration on Cisco Router with connected clients (192.168.1.0/24 network)

6. Discussion

The hub-and-spoke WAN design kept the routing configuration simple and gave both branches a single, predictable path to the centralized services, though it does mean the Server Site link is a single point of failure for DHCP, HTTP and DNS if it goes down - a limitation that would be worth addressing in a production deployment with a backup link or a secondary server. RIP was an appropriate choice for a network this size because its configuration is minimal and convergence was fast in testing, but it would not scale well if the company added many more branches, since RIP's hop-count metric and slower convergence become limiting factors in larger topologies; OSPF would be a more future-proof choice at that point. The decision to relay DHCP requests across the WAN, rather than placing a DHCP server in each branch, matches the project's centralization goal but adds a dependency on the WAN link and on ip helper-address being configured correctly on every department interface - a single missing helper-address would silently break DHCP for that department only, which is a subtle failure mode worth documenting for whoever administers the network later. Isolating Finance in its own VLAN with static addressing achieved the intended access control, though it also means new Finance hosts must be configured manually, a reasonable trade-off given the sensitivity of that department's data.

7. Conclusion

The project successfully designed, implemented and verified a multi-branch company network in Cisco Packet Tracer that meets every stated requirement: two branches and a remote server site connected by RIP, five departments per branch addressed through a mix of dynamic, wireless and static strategies, VLAN isolation and inter-VLAN routing for the finance department, and centralized DHCP, HTTP and DNS services reachable from both branches. Testing confirmed correct address allocation, wireless connectivity, VLAN isolation, RIP route propagation and reachability of the centralized services from every department, giving the organization a working network foundation that reflects real-world practice for a small, geographically distributed company.

8. References

- [1] Cisco Networking Academy, Routing and Switching Essentials Companion Guide, Cisco Press.
- [2] Cisco Packet Tracer software documentation, Cisco Systems Inc.

9. Appendices

Abbreviations

- DHCP - Dynamic Host Configuration Protocol
- DNS - Domain Name System
- HTTP - Hypertext Transfer Protocol
- RIP - Routing Information Protocol
- VLAN - Virtual Local Area Network
- VLSM - Variable-Length Subnet Masking
- WAN - Wide Area Network
- WLAN - Wireless Local Area Network